AN ANALYSIS OF TRENDS IN TRAVEL AND PUBLIC TRANSPORT ACCESSIBILITY IN ENGLAND (2002-2023)

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Module: Principles of Data Presentation

Assignment: Mini group project

GitHub link:

https://github.com/aisha2528/Mini-Group-Project

[Individual Section]

Abstract

This analysis focuses on understanding the evolution of travel behaviour in England between 2002 and 2023, in the context of rural and urban variations. Applying data from National Travel Survey NTS9903 data, the investigation seeks to examine the changes in travel behaviours in terms of private car ownership, dependence on public transportations, and walking activities. Based on data cleaning, metrics development, and visualisations, performed via Python, the comparison is made in relation to baseline values in 2002 and 2023. According to the outcomes of the analysis, there is an increasing gap between rural and urban areas in using buses, a declining trend in terms of trips per capita across the country, and an inverse correlation between walking and driving in urbanised areas.

[Group Section]

Introduction & Background

Transport systems serve as the backbone of social and economic participation, providing essential links to work, education, and healthcare. over the last twenty years, England has witnessed a transformation in travel behaviour driven by digital technology, shifting work patterns, and evolving environmental standards. However, access to these systems is not uniform across the country.

This report explores the trends in travel modes across different geographic classifications. By comparing Urban conurbations with rural hamlets, we aim to identify where public transport is succeeding and where private vehicle dependency remains entrenched. We used the National Travel Survey (NTS) as our primary data source, which provides a comprehensive longitudinal look at how residents of England move in their daily lives.

[Group Section]

Methods

The analysis was conducted using the NTS9903 dataset (Rural-Urban classification of residence). The technical workflow was managed through a Python script using the Pandas library for data manipulation and

Matplotlib for visualisation. We focused on a comparison between the 2002/03 baseline and the 2023 data to capture long-term structural changes.

Data cleaning involved converting categorical identifiers and managing missing or low-value entries (marked as '[low]' in the original set) to ensure numerical accuracy. Three primary visualisations were generated: a bar chart for geographic mode comparison, a line chart for temporal trends, and a scatter plot to analyse the correlation between active travel and driving.

[Group Section]

Results

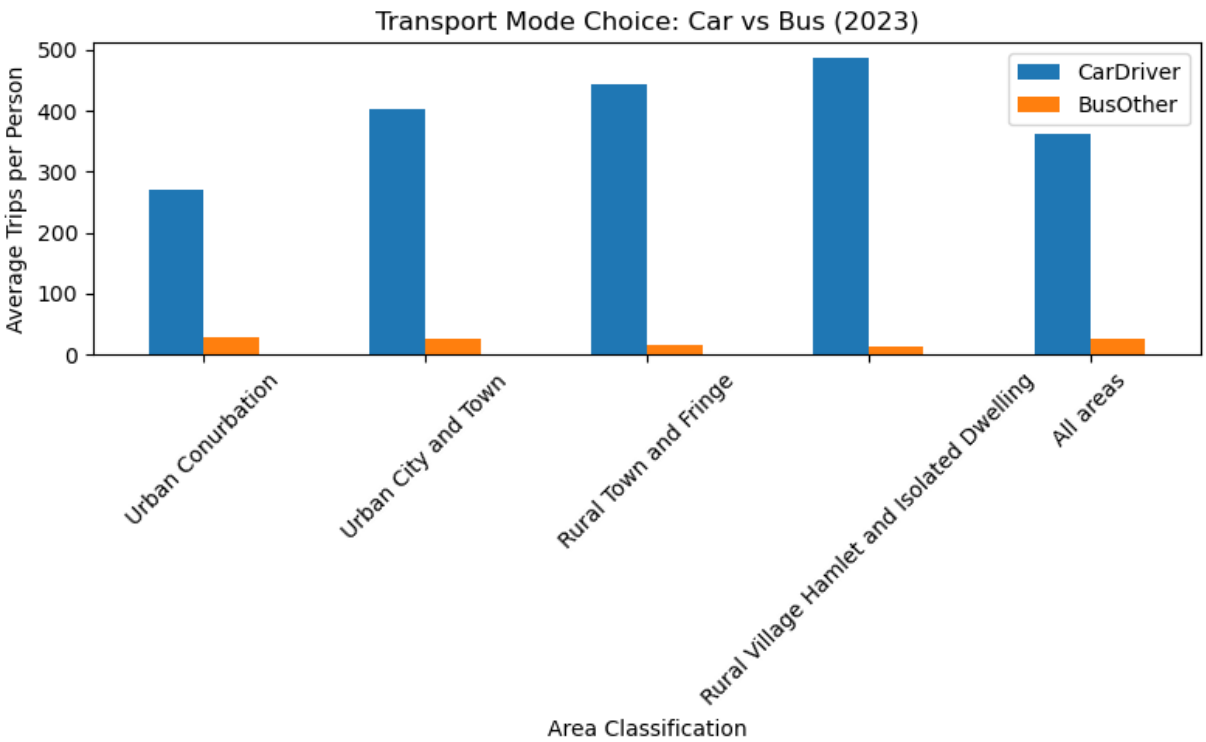

Figure 1: Comparison of Car vs Bus trips by Area Classification (2023)

The results indicate a stark contrast in transport reliance based on residence type. Figure 1 illustrates that in 2023; rural residents made significantly more trips as car drivers (487 per year) compared to those in urban conurbations (271 per year).

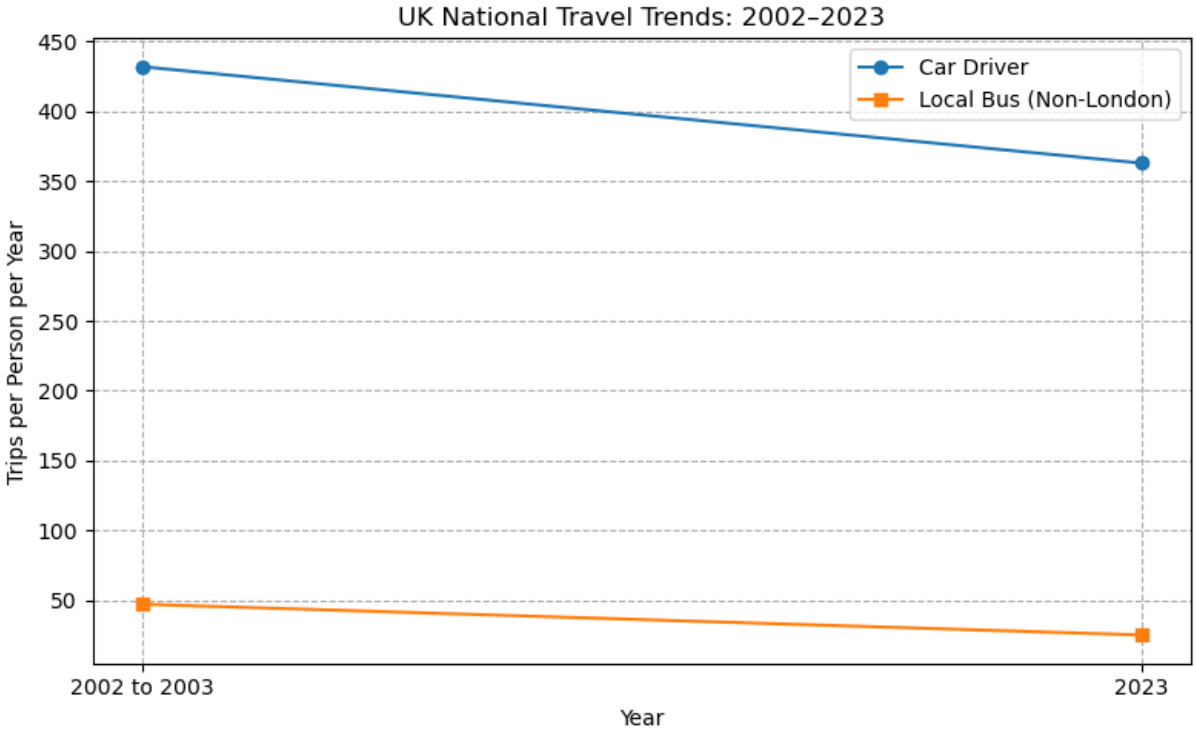

Figure 2: England National Travel Trends from 2002 to 2023

Furthermore, Figure 2 shows the national trend over two decades. While the total number of trips per person has trended downwards, the gap between private and public transport use remains wide.

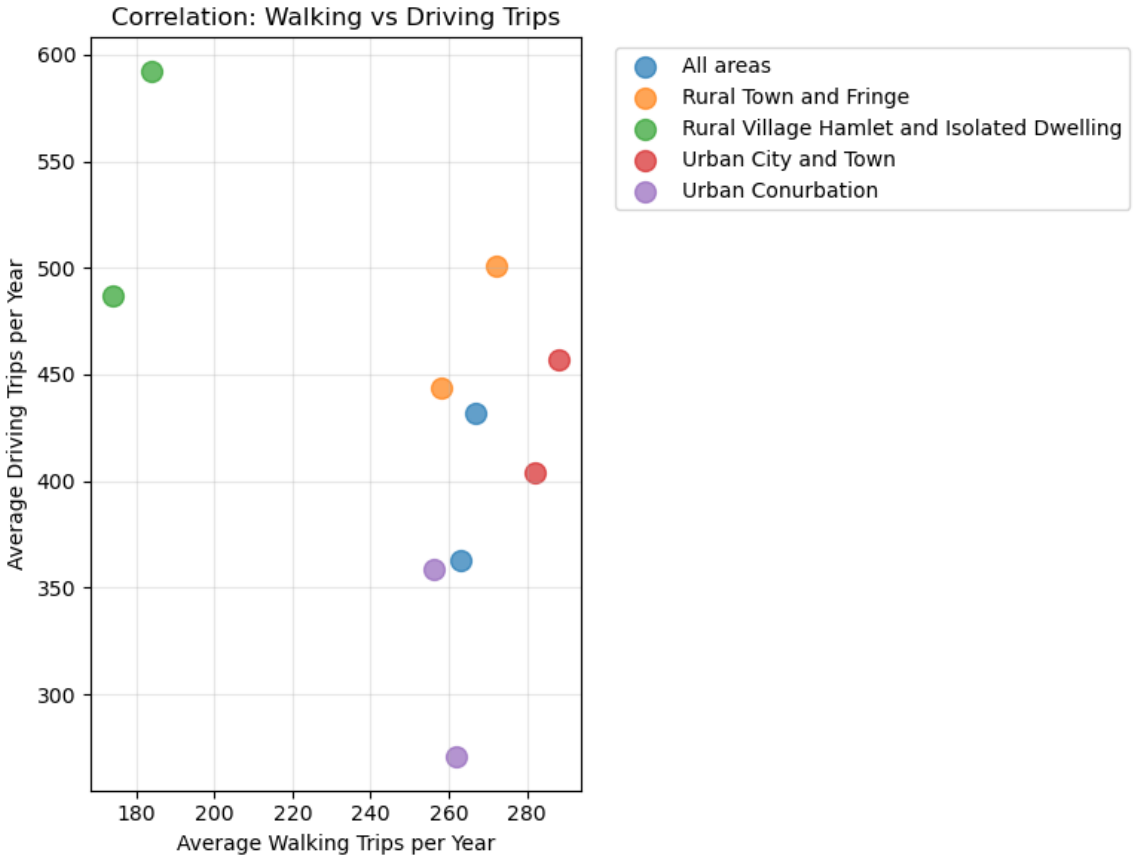

Figure 3: Correlation between Walking and Driving Frequencies

Finally, Figure 3 explores the relationship between walking and driving. Clustering suggests that urban

infrastructure that promotes walking successfully correlates with reduced car dependency.

[Individual Section]

Discussion

This information shows that the transport system in England is extremely unbalanced geographically. The phenomenon known as the 'rural-urban divide' is not determined by the choice to use a car but rather reflects serious disparities in transport infrastructure development. Living in cities, citizens can comfortably rely on numerous public transport options such as buses and trains. For rural areas, however, it is hard to imagine an alternative to using one's own car.

The main trend that can be observed in the line chart is that the number of trips decreased after 2002. This phenomenon should be viewed in the context of digital transformation as people started teleworking, conducting bank transactions, and buying goods online. Nevertheless, such changes do not benefit rural areas where it is necessary to visit various institutions including health care facilities.

One of the weaknesses of the research is its dependence on statistics collected through the National Travel Survey. The problem is that the method under discussion shows national figures whereas it is important to focus on local communities to understand that transport poverty still exists.

[Individual Section]

Conclusion

The analysis of the data set of NTS9903 reveals that there is significant evidence of transformation in travel behaviour in England over the last twenty years; however, these changes do not affect all groups in society equally. People living in urban centres still use public transport extensively; at the same time, rural citizens depend on car use due to the reduced presence of buses. Overall, a decline in the number of trips made annually can be seen from the national perspective, which suggests that there is a shift in lifestyle caused by such phenomena as digitalisation and working remotely. Moreover, the evident inverse association between walking and car use in urban areas confirms the necessity of efficient infrastructure development to encourage sustainable transportation.

[Group Section]

References

Department for Transport (2024). *National Travel Survey: 2023*. [Online] Available at:

www.gov.uk/government/statistics/national-travel-survey-2023